

TITLE OF PROJECT: Student-Generated Educational Games in CS: A Constructionist Approach to AI-Assisted Prototype Development

NAME OF PRINCIPAL INVESTIGATOR(S): Kathleen Kelly

CONTACT NAME AND EMAIL FOR QUESTIONS/PROBLEMS: Kathleen Kelly, kathleenmariekelly@mines.edu

RESEARCH DESCRIPTION

What the study is about

The purpose of this study is to research what advanced Computer Science (CS) students learn about pedagogy and their own CS knowledge by designing games that teach CS concepts, and whether structured scaffolding of AI-assisted design changes how students reason about educational design. As a part of research for a PhD, Kathleen is studying computer science education at varying age levels. The purpose of this study is to research what advanced CS students learn about pedagogy and their own CS knowledge by designing games that teach CS concepts. Students will come up with an educational game concept, then using AI as a partner, iterate on the idea of the game, create a prototype, and conclude with playtesting of each other's games. This study is investigating how undergraduate students approach the design of educational games, and how that design thinking evolves over the course of an AI-assisted iteration process. Student memos, reflection, and AI transcripts will be analyzed as part of this study. All data will be deidentified before analysis.

For this study, Kathleen is comparing two different ways of iterating on an educational game design with AI. Some students will receive a structured guide Kathleen developed; others will iterate using ordinary prompting. Both are reasonable approaches, and Kathleen is genuinely uncertain which produces better design thinking, hence this research project. You as students have the job of engaging honestly with the process.

What will you be asked to do

Students will be asked to submit the results of this educational game activity to Kathleen to use in a research study and publication. This activity will be given to consenting students on a Saturday. Participation in the research study for Kathleen's paper will be voluntary.

Students will:

1. **Write a short pre-design memo** describing your game and your reasoning behind it, before beginning structured iteration.
2. **Iterate on your design using an AI assistant** over approximately one hour. Some students will receive a structured design guide (a "skills" document) to use during their AI iteration; others will iterate without the guide. Assignment to these two approaches will be random. Both approaches are legitimate, and all students will have access to both after the study concludes.
3. **Write a second post-design memo** describing your design after iteration.
4. **Write a brief reflection** on how your thinking about the design changed over the course of iteration.
5. **Submit your entire project zip file** into a shared drive.
6. **Conduct a playtesting session** with peers and submit the resulting data.

The memos, reflection, transcripts, prototype, and playtest data will be used for research analysis. You will complete these activities as part of the one day event.

If you consent to participate, you will be randomly assigned to one of two groups:

- **Group A:** You will receive a structured design guide ("skills" document) to use during your AI iteration.
- **Group B:** You will iterate with AI using ordinary prompting, without the structured guide.

Assignment is determined by a random procedure and cannot be changed. Both groups complete the same course assignment with the same deadlines and expectations. After the study concludes, all students will have access to the structured design guide.

Potential risks

Students should not feel any discomfort during this study, as it is a voluntary event. Participation in research is completely voluntary and will not count towards students grades in any way.

Potential benefits

Students taking part in this experiment will benefit the overall study of educational games for undergraduate computer science students. Student performance in the experiment as well as feedback will be greatly beneficial to this research project and potential additions to course structures in the future.

PARTICIPANT'S RIGHTS

Confidentiality

Student information collected will be anonymized. Students will be submitting their game design document, their interaction with AI, their prototype, their playtesting data, and a reflection. Students may also be asked follow-up questions in an interview related to how the experience helped or hindered the student. Any student data referenced in future publications will also be anonymized.

Information obtained about you for this study will be treated as confidential to the extent allowed by law. Research information that identifies you may be shared with the Colorado School of Mines Human Subjects Team and others who are responsible for ensuring compliance with laws and regulations related to research.

Data rights

You have rights around the protection and movement of your data. This consent form provides transparency to the data collected about you and how it will be used. Because we are not collecting any identifying information about you, it will not be possible to extract your individual data if you want to withdraw your data from the study.

You have rights around the protection and movement of your data. This consent form provides transparency to the data collected about you and how it will be used. Your rights include (but not limited to):

- Right to correct inaccurate personal data;
- Right to erase data that is no longer necessary for the research purposes or that you withdraw your consent to use. Note: if de-identified data is used, once identifiers are removed from the data and the data is added to the data set to be analyzed, it will be impossible to extract your individual data if you want to withdraw your data from the research.
- Right to know whether your personal data has been processed, what's been processed, and the purpose for such.
- Right to restrict processing of the data where it's inaccurate, unlawfully processed, or is no longer needed.
- Right to object to processing personal data for the purposes of public interest or legitimate interest pursued by the research and/or Mines.

Voluntary participation and withdrawal

Your participation in this research is voluntary. You have the right to withdraw from the study at any time without consequence. To withdraw, inform the principal investigator of your intention to withdraw.

If you are a Mines student or employee, taking part in this study is not part of your class work or duties. You can refuse to participate, or withdraw at any time, with no effect on your class standing, grades, or employment at Mines.

CONTACT

Whom to contact if you have questions about the study: If you have questions, concerns, or complaints about this research, its procedures, risks and benefits, contact the principal investigator at kathleenmariekelly@mines.edu.

Independent contact: If you have questions about your rights as a research participant or are not satisfied with how this study is being conducted, please contact the Colorado School of Mines Human Subjects Research committee at humansubjects@mines.edu.

Your signature acknowledges that you have read the information stated and willingly sign this consent form. Your signature also acknowledges that you have received, on the date signed, a copy of this document containing 2 pages.

Travis Cox P41

Participant name (printed)


Participant signature

9/19/2026

Date

If you prefer not to participate in this research study, you may sign below instead and be excused.

Participant name (printed)

Participant signature

Date

I decline to participate in the research study.